Centralia
TECHNOLOGY

Capture The Flag

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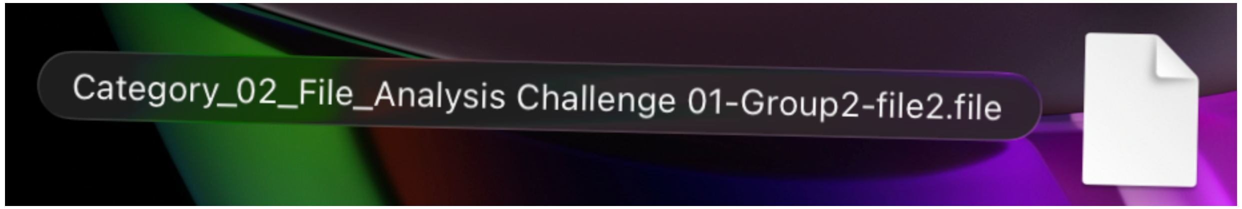
Introduction

- The CTF Problem
- Steps to Solve
- The Solution
- Workplace Relevance

CTF Category Description



- I chose the Category 2 File Analysis challenge.

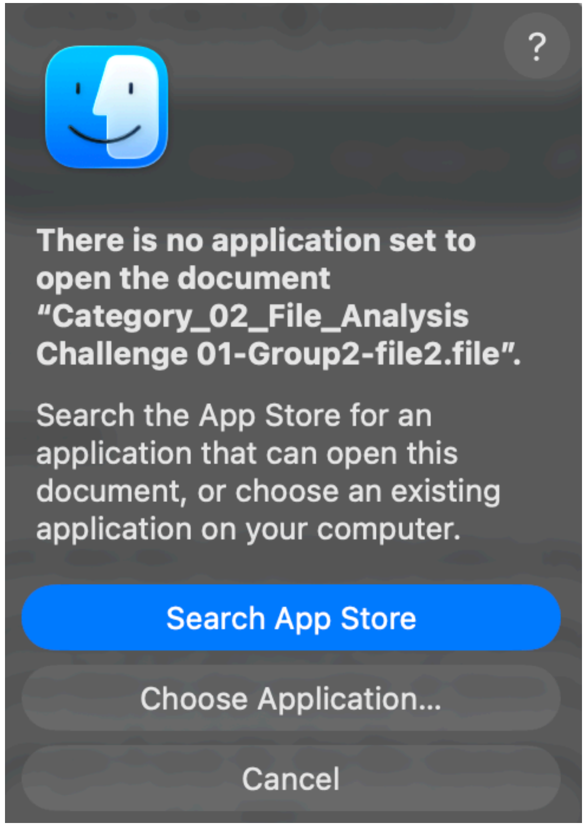
  This challenge involves finding out what the filename and extension are for a corrupted file.

Introduction to the Problem

Upon trying to open the file, I was met with this screen. My computer didn't know what to do with this file.

When attempting to print the file using the cat command, mere gibberish is spat out.

It's time to figure out what this file is based off the magic bytes; a file's Metadata.



Download exiftool

```
[$ exiftool -ver
zsh: command not found: exiftool
$ 
```

A great tool to use here is a parser like exiftool.

Exiftool works by reading the metadata embedded in the file, and comparing it to known signatures to identify the file and its contents

```
[$ brew install exiftool
==> Auto-updating Homebrew...
Adjust how often this is run with `$HOMEBREW_AUTO_UPDATE_SECS` or disable with
`$HOMEBREW_NO_AUTO_UPDATE=1`. Hide these hints with `$HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
==> Auto-updated Homebrew!
Updated 2 taps (homebrew/core and homebrew/cask).
==> New Formulae
erfa: Essential Routines for Fundamental Astronomy
floresta: Lightweight and embeddable Bitcoin client, built for sovereignty
forgecode: AI-enhanced terminal development environment
hf-mount: Mount Hugging Face Buckets and repos as local filesystems
lazymake: Modern TUI for Makefiles
mcp-remote: Remote proxy for Model Context Protocol with OAuth support
openssl@4: Cryptography and SSL/TLS Toolkit
rabbitmqadmin: Command-line tool for RabbitMQ that uses the HTTP API
sdl3_mixer: Sample multi-channel audio mixer library
try: Quickly manage and navigate project directories for experiments
vite-plus: Unified toolchain and entry point for web development

You have 11 outdated formulae installed.

==> Fetching downloads for: exiftool
✔ Bottle Manifest exiftool (13.55)
  Downloaded   19.3KB/ 19.3KB
✔ Bottle exiftool (13.55)
  Downloaded   16.7MB/ 16.7MB
==> Pouring exiftool--13.55.arm64_tahoe.bottle.tar.gz
🍺  /opt/homebrew/Cellar/exiftool/13.55: 1,276 files, 57MB
==> Running `brew cleanup exiftool`...
Disable this behaviour by setting `HOMEBREW_NO_INSTALL_CLEANUP=1`.
Hide these hints with `HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
$ 
```

Arriving at the Solution

- All it takes is a single command; exiftool <filename>

```
$ exiftool [redacted] Category_02_File_Analysis\ Challenge\ 01-Group2-file2.file
ExifTool Version Number         : 13.55
File Name                       : Category_02_File_Analysis Challenge 01-Group2-file2.file
Directory                       [redacted]
File Size                       : 2.0 MB
File Modification Date/Time     : 2026:04:26 13:10:45+02:00
File Access Date/Time           : 2026:04:26 13:10:45+02:00
File Inode Change Date/Time     : 2026:04:26 13:10:46+02:00
File Permissions                : -rw-r--r--
File Type                       : Win32 EXE
File Type Extension             : exe
MIME Type                       : application/octet-stream
Machine Type                    : Intel 386 or later, and compatibles
Time Stamp                      : 2015:10:28 00:10:55+01:00
Image File Characteristics      : Executable, Large address aware, 32-bit
PE Type                         : PE32
Linker Version                  : 12.0
Code Size                       : 1403392
Initialized Data Size           : 687104
Uninitialized Data Size         : 0
Entry Point                     : 0x1333f8
OS Version                      : 5.1
Image Version                   : 0.0
Subsystem Version               : 5.1
Subsystem                       : Windows GUI
File Version Number             : 4.21.0.0
Product Version Number          : 4.21.0.0
File Flags Mask                 : 0x003f
File Flags                      : (none)
File OS                         : Windows NT 32-bit
Object File Type                : Executable application
File Subtype                    : 0
Language Code                   : English (U.S.)
Character Set                   : Unicode
Company Name                    : Sysinternals
File Description                : BGInfo - Wallpaper text configurator
File Version                    : 4.21
Internal Name                   : BGInfo
Legal Copyright                 : Copyright © 2000-2014 Mark Russinovich
Original File Name              : Bginfo.exe
Product Name                    : BGInfo
Product Version                 : 4.21
```

Strategies, Pitfalls, Lessons Learned

- The strategy employed was a simple one that revealed the original filename and extension as well as a myriad of other information.
- When deciding upon a strategy, I also considered using the “strings <filename>” command, but decided against using it unless exiftool failed.
- I learned the difference between a parser like exiftool and a scavenger like strings.

The Relationship to the Workplace

- File analysis is an essential tool for identifying lost, corrupted, or suspicious files in the workplace. Take an SOC Analyst role for example.

  Perhaps a file was downloaded but flagged by an antivirus and quarantined. Upon using the strings command, you find a URL exfiltrating data, hiding deep in the application, only visible when examining the binary through brute force.

  Perhaps a file was recovered from a corrupted drive and has lost some of its data. Using exiftool, you can parse the metadata in a way that's readable and can aid in a file's recovery.

Summary

- In this presentation, I analyzed a corrupted file that was unrecognizable to my laptop, utilized exiftool to extract the metadata, and successfully identified the file name and extension.
- The core takeaway of this presentation is the demonstrated importance of file analysis in digital forensics, incident response, and data recovery.

References

[1] P. Harvey, "ExifTool by Phil Harvey," *ExifTool.org*. [Online]. Available: https://exiftool.org/. [Accessed: Apr. 26, 2026].

[2] "exiftool — Homebrew Formulae," *Brew.sh*. [Online]. Available: https://formulae.brew.sh/formula/exiftool. [Accessed: Apr. 26, 2026].